

Lightweight CNN-Based Object Classification on CIFAR-10 with Classical ML Comparison

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Abstract—This paper presents a compact convolutional neural network (CNN) designed for image classification on the CIFAR-10 dataset. The proposed model consists of five convolutional layers grouped into three convolutional blocks, followed by two fully connected layers with dropout regularization. The network contains approximately 666,000 trainable parameters, making it significantly smaller than many commonly used deep architectures.

The model was trained for 10 epochs on an NVIDIA T4 GPU using Google Colab, requiring approximately 4 minutes of training time. It achieved a training accuracy of 80.39% and a test accuracy of 77.59%.

To better understand its effectiveness, the proposed CNN was compared with classical machine learning approaches including Support Vector Machine (SVM), K-Nearest Neighbors (KNN), and Random Forest classifiers trained on flattened pixel inputs. Transfer learning using ResNet18 was also evaluated for reference. Results show that the proposed lightweight architecture provides a strong balance between accuracy and computational cost, while classical models perform substantially worse on raw pixel representations.

Index Terms—CIFAR-10, Convolutional Neural Network, Transfer Learning, Image Classification, Classical Machine Learning

I. INTRODUCTION

Image classification remains one of the most studied problems in computer vision. While traditional machine learning models rely on handcrafted or flattened features, convolutional neural networks (CNNs) are capable of learning hierarchical spatial representations directly from images.

Deep architectures such as ResNet have demonstrated excellent performance, but they often require millions of parameters and significant computational resources. In practical scenarios, especially in resource-constrained environments, smaller models are desirable.

In this work, we design a lightweight CNN architecture for CIFAR-10 classification and compare its performance with classical machine learning models and transfer learning using ResNet18.

II. CONTRIBUTIONS

The main contributions of this paper are:

- Design and implementation of a lightweight CNN architecture with approximately 666,000 parameters.
- Mathematical formulation of convolution, activation, and optimization used in the model.

- Experimental comparison with classical ML models and transfer learning.
- Analysis of accuracy versus computational efficiency.

III. DATASET DESCRIPTION

The CIFAR-10 dataset [1] contains 60,000 color images of size 32×32 across 10 object categories. The dataset is divided into 50,000 training images and 10,000 test images. Each class contains an equal number of samples, making the dataset balanced.

IV. MATHEMATICAL FORMULATION

A. Convolution Operation

For an input tensor $X \in R^{H \times W \times C}$, convolution with a filter W is defined as:

$$Y_{i,j}^{(k)} = \sum_{c=1}^C \sum_{m=1}^K \sum_{n=1}^K W_{m,n,c}^{(k)} X_{i+m,j+n,c} + b^{(k)}$$

where $W^{(k)}$ denotes the k -th learnable kernel and $b^{(k)}$ is the bias.

B. Activation Function

The Rectified Linear Unit (ReLU) introduces non-linearity:

$$f(x) = \max(0, x)$$

C. Loss Function

The final layer produces logits z_i , which are converted into probabilities using the Softmax function:

$$\hat{y}_i = \frac{e^{z_i}}{\sum_{j=1}^{10} e^{z_j}}$$

The categorical cross-entropy loss is defined as:

$$\mathcal{L} = - \sum_{i=1}^N \sum_{c=1}^{10} y_{i,c} \log(\hat{y}_{i,c})$$

Optimization is performed using the Adam algorithm.

Proposed Lightweight CNN Architecture for CIFAR-10

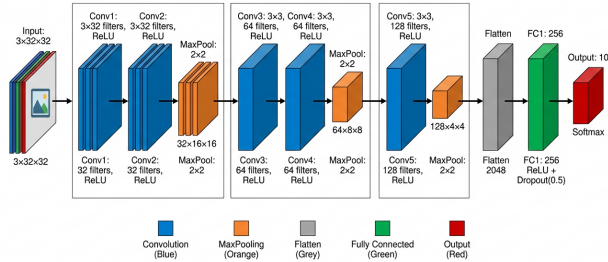


Fig. 1. Proposed Lightweight CNN Architecture.

V. PROPOSED LIGHTWEIGHT CNN ARCHITECTURE

The proposed network consists of three convolutional blocks followed by fully connected layers.

Each convolutional block contains one or two convolution layers followed by ReLU activation and max pooling. After the final pooling layer, feature maps are flattened and passed through a fully connected layer of 256 neurons with dropout (0.5), followed by a 10-class output layer.

The total number of trainable parameters is approximately 666,000.

VI. EXPERIMENTAL SETUP

The model was trained under the following configuration:

- Optimizer: Adam
- Learning Rate: 0.001
- Batch Size: 64
- Epochs: 10
- Hardware: NVIDIA T4 GPU (Google Colab)

Data augmentation included random cropping and horizontal flipping.

VII. COMPUTATIONAL EFFICIENCY

ResNet18 [2] contains approximately 11 million parameters, whereas the proposed network uses fewer than one million. Training required approximately 4 minutes on a T4 GPU for 10 epochs. The reduced parameter count lowers memory usage and training time while still achieving competitive accuracy.

VIII. RESULTS AND PERFORMANCE COMPARISON

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Test Accuracy (%)
KNN	28.97
SVM	32.44
Random Forest	42.16
Lightweight CNN	77.59
ResNet18	89.27

The proposed CNN achieved a training accuracy of 80.39% and a test accuracy of 77.59%. The macro-averaged F1-score

Training Loss Curve

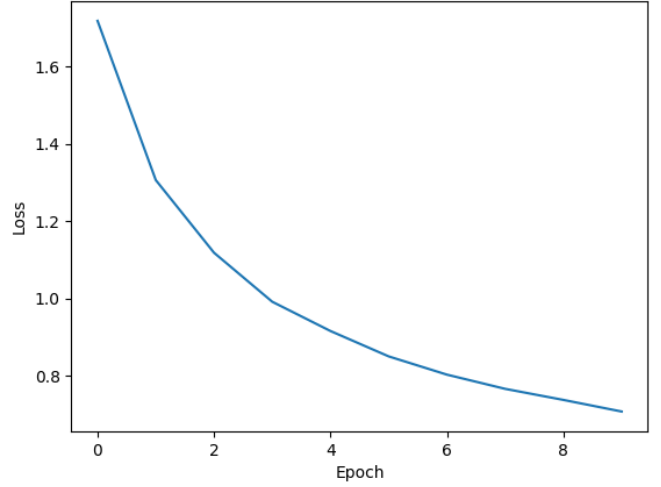


Fig. 2. Training loss curve over 10 epochs.

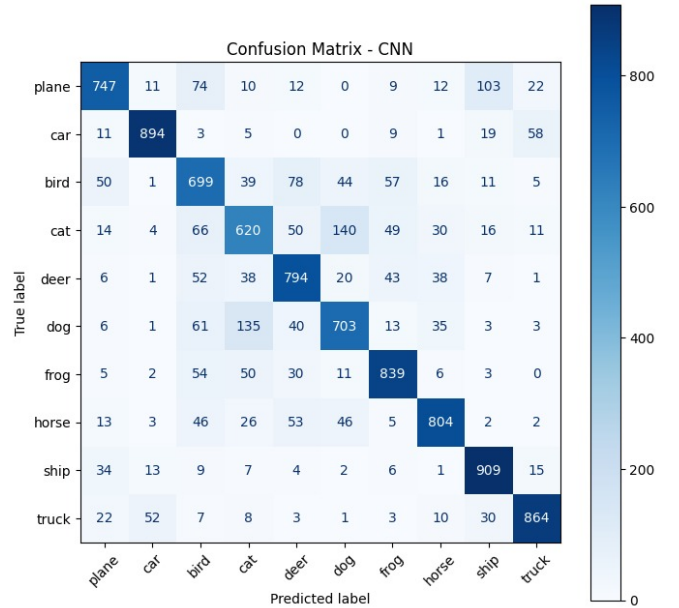


Fig. 3. Confusion matrix of the proposed CNN on CIFAR-10 test set.

was 0.78, which closely matches overall accuracy due to the balanced nature of the dataset.

Analysis of the confusion matrix shows that vehicle categories such as automobile and truck are classified with higher accuracy, likely due to clearer structural differences. Greater confusion is observed among visually similar animal classes such as cat and dog, which is common in CIFAR-10 due to low image resolution.

IX. FILTER VISUALIZATION

Visualization of the first convolutional layer shows edge-oriented and color-gradient patterns, indicating that the network learns meaningful low-level features.



Fig. 4. Visualization of first-layer convolutional filters.

X. DISCUSSION

The results suggest that even a moderately deep CNN with limited parameters can achieve reasonable performance on CIFAR-10. While deeper pretrained models provide higher accuracy, the proposed architecture offers a trade-off between performance and efficiency.

Classical machine learning models performed poorly on flattened pixel inputs, highlighting the importance of spatial feature learning in image classification.

XI. LIMITATIONS

Several limitations should be noted:

- The model performs below deeper pretrained architectures such as ResNet18.
- No robustness or adversarial testing was conducted.
- Data augmentation strategies were limited.
- Cross-dataset generalization was not evaluated.
- Batch normalization was not included in the architecture.

XII. CONCLUSION

A lightweight CNN architecture was developed for CIFAR-10 classification and evaluated against classical ML methods and transfer learning. The model achieved 77.59% test accuracy with relatively low computational cost. While transfer learning provided higher accuracy, the proposed model demonstrates that compact architectures can still perform competitively.

Future work may include stronger augmentation, normalization layers, and further architectural refinements.

ACKNOWLEDGMENT

The authors thank Dr. P. Mirunalini, Dr. J. Bhuvana, and Dr. Poreddy Ajay Kumar Reddy, Department of Computer Science, Sri Sivasubramaniya Nadar College of Engineering, for their valuable guidance and support throughout this work.

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